

Embedding and Contingent Homomorphisms

Syntax, semantics, acquisition, diversity, morphology, and phonology

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Embedding preserves structure across distinct domains

Question

Which correspondences between form and meaning are necessary, and which are contingent facts of a language?

- Syntax and semantics organize dependencies through embedding.
- A homomorphism preserves selected composition across domains.
- Linguistic correspondences remain empirical: possible mathematics does not select the actual grammar.

MOLR §§6.1–6.4

Where we are

1. Embedding
2. Homomorphism
3. Acquisition and diversity
4. Nonembedding

Syntactic embedding is structural containment

- ① The man [the dog [the cat chased] bit] laughed.
 - ② I know [you know [she knows [John believes [the earth is flat]]]].
- One admissible expression occupies a position inside another.
 - The dependency is recursive as a description, finite in each token.
 - Linear adjacency neither guarantees nor prevents embedding.

MOLR §6.1

Semantic embedding composes functions and arguments

$$\llbracket X Y \rrbracket = \llbracket X \rrbracket(\llbracket Y \rrbracket) \quad \text{or} \quad \llbracket Y \rrbracket(\llbracket X \rrbracket)$$

when the relevant types connect.

- Composition must be defined on the values supplied.
- Type notation records admissibility, not the meaning itself.
- Structural combination and semantic combination must correspond somehow, but not necessarily identically.

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A homomorphism preserves an operation

For structures A and B , mapping $h : A \rightarrow B$ is homomorphic when

$$h(x \star_A y) = h(x) \star_B h(y).$$

- The domains and operations may be different.
- What is preserved must be stated.
- The mapping need not be one-to-one or onto.

Question

What would a syntactic concatenation–semantic composition homomorphism preserve?

MOLR §6.2

Linguistic homomorphisms are contingent

- Mathematics supplies many structure-preserving mappings.
- A language instantiates only some of them.
- Lexical categories and constructions determine which compositions are admitted.
- Cross-linguistic variation reveals alternatives; acquisition selects a locally sustained system.

Take-away

The homomorphism is not axiomatic merely because it can be written mathematically.

Admissibility filters possible form–meaning pairings

possible mappings $\supset$ structurally admitted $\supset$ language-particular $\supset$ contextually successful

- Composition provides possibilities.
- Grammar constrains category connections.
- Lexical and constructional facts select language-particular correspondences.
- Use in activity adds further conditions.

Aspect shows why lexical correspondence matters

- ① *She won the competition for an hour.
 - ② She won the competition for a year.
 - ③ She played the piano in a year.
 - ④ *She played the piano in an hour.
- The same surface modifier combines with distinct event structures.
 - Successful composition depends on the semantic value assigned to the verb construction.

Guided analysis

Propose a coerced reading that repairs each starred example.

Where we are

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Contingency links the three fundamental questions

- ① Children seek usable correspondences among structured signals, situations, and actions.
- ② Continuity arises where child systems preserve and refine such correspondences.
- ③ Diversity arises because several mappings can preserve shared dependency types.

Take-away

Limited kinds of dependency can coexist with extensive variation in their realization.

MOLR §6.3

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Morphology and phonology can decompose without embedding

- A word or phonological form can be analyzed into ordered parts.
- Decomposition does not entail that each part contains another expression of the same kind.
- Local relations may be regular or linear without becoming syntactic embedding.

Question

Why is internal complexity not by itself evidence of recursion?

MOLR §6.4

Turkish compounds show structured decomposition

Kadıköy vergi daire-si
Kadıköy tax office-POSS.3SG

- The possessive marking participates in compound structure.
- Not every imaginable bracketing or substring is independently admissible.
- The analysis must distinguish morphological constituency from free syntactic embedding.

Guided analysis

Compare the interpretations predicted by the two binary bracketings.

Third hour: test preservation rather than resemblance

Choose one syntactic construction and its semantic analysis.

- ① State the source objects and operation.
- ② State the target objects and operation.
- ③ Define the proposed mapping.
- ④ Test one preserving and one failing case.
- ⑤ Identify which facts make the mapping language-contingent.

Week 11 synthesis

- ① Embedding concerns structural containment, not mere length or complexity.
- ② Homomorphisms preserve specified operations across domains.
- ③ Linguistic homomorphisms are selected empirically, not guaranteed axiomatically.
- ④ Contingency links acquisition, continuity, and diversity.
- ⑤ Decomposition in morphology and phonology need not constitute embedding.

Question

Quiz 9 asks you to state or test one proposed structural correspondence.